CONTENTS :

- IMPORTANCE OF WATER
- OBJECTIVES OF PROTECTED WATER SUPPLY
- COMPONENTS OF A PROTECTED WATER SUPPLY SCHEME
- PERCAPITA DEMAND
- SYSTEMS OF DISTRIBUTION

General Importance of water

- No life can exist without water
- It is as essential for life as air is
- Two-third of human body is constituted of water
- Provides luxuries and comforts to human beings

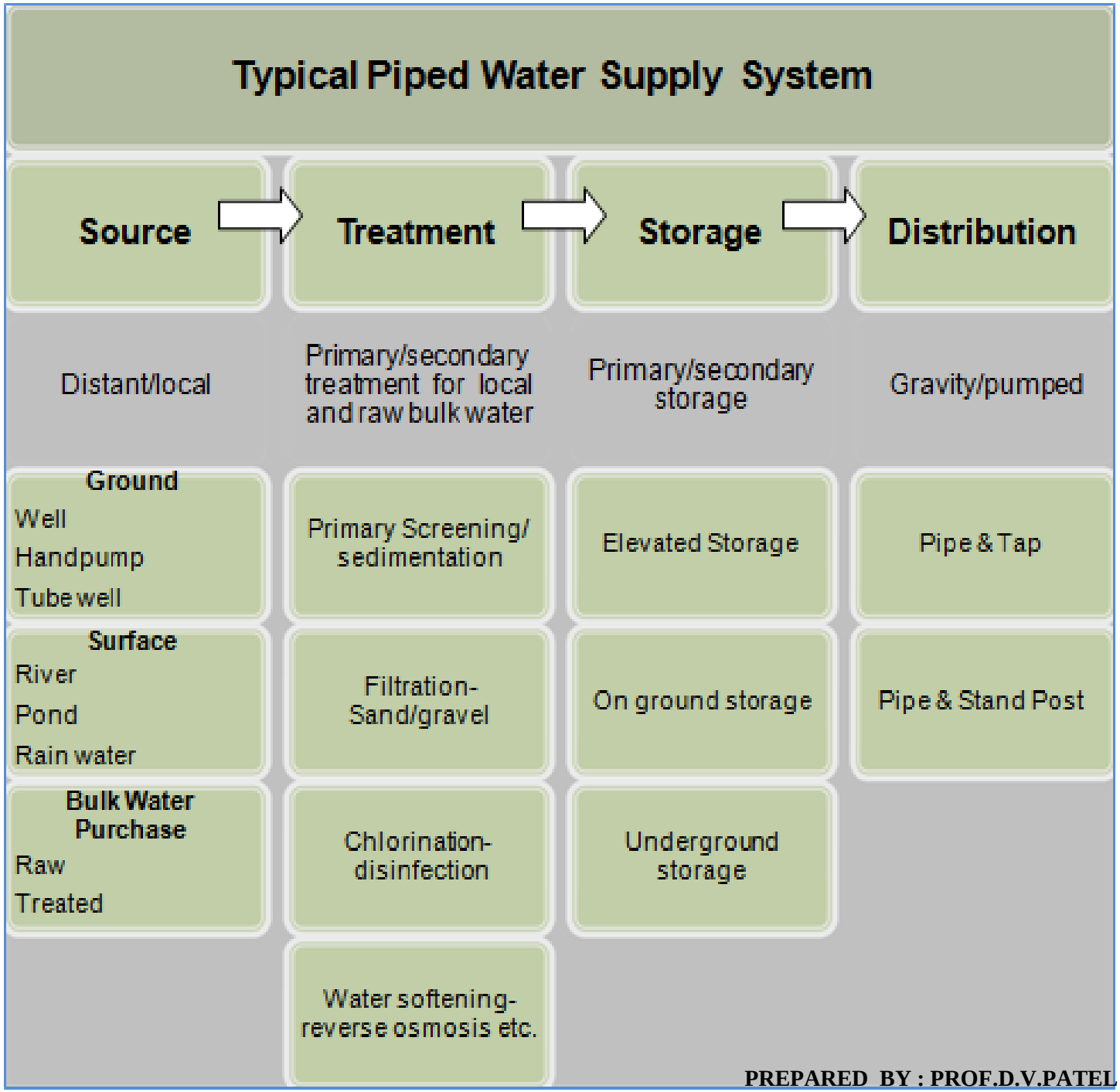
Objectives of protected water supply

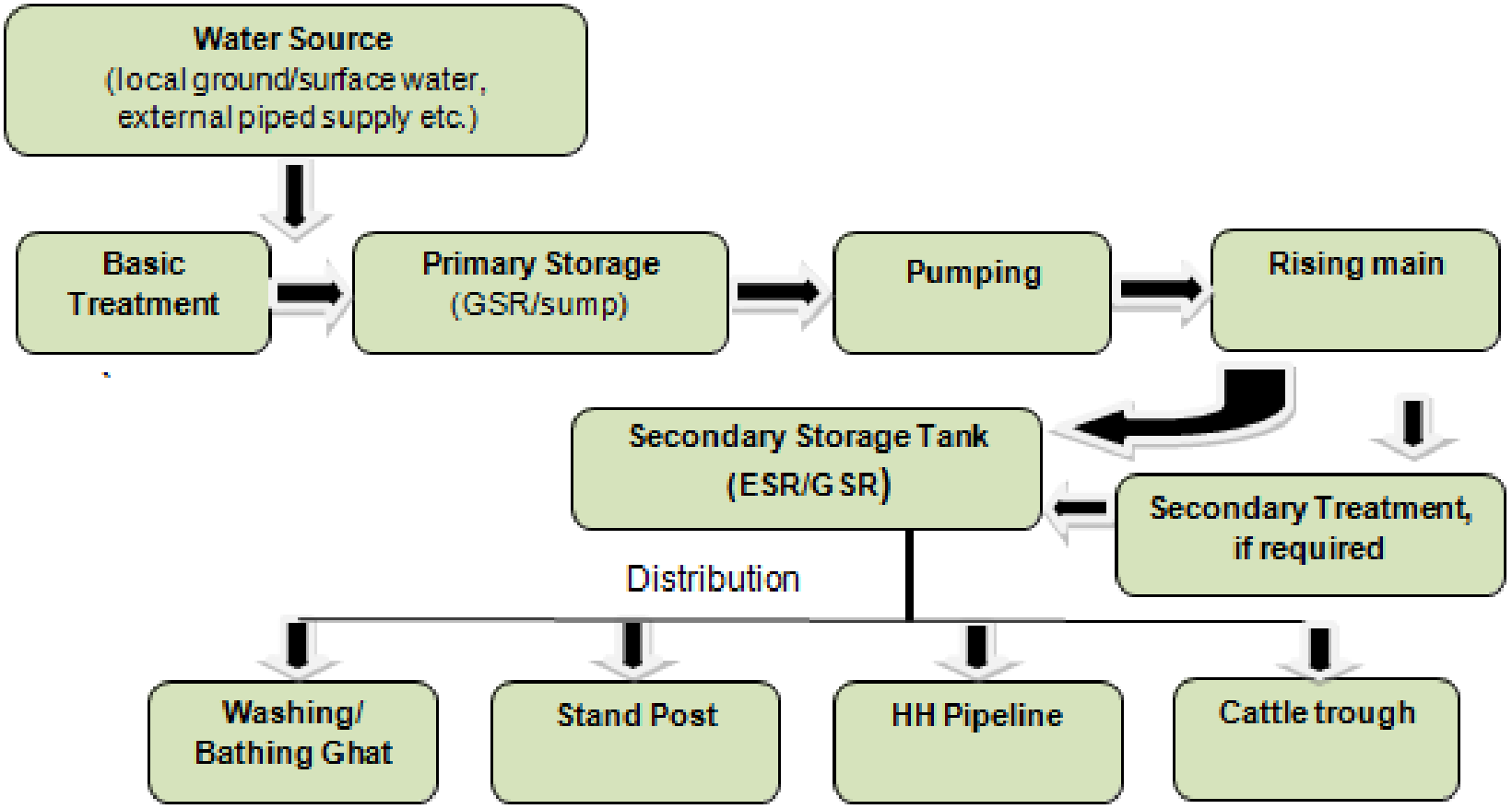
The objectives of protected water supply scheme are

- To supply safe and wholesome water to consumers
- To supply water in sufficient quantities
- To supply water at convenient points and timings
- To supply water at reasonable cost to the users
- To encourage personal and house hold cleanliness of users

Typical Water Supply System:

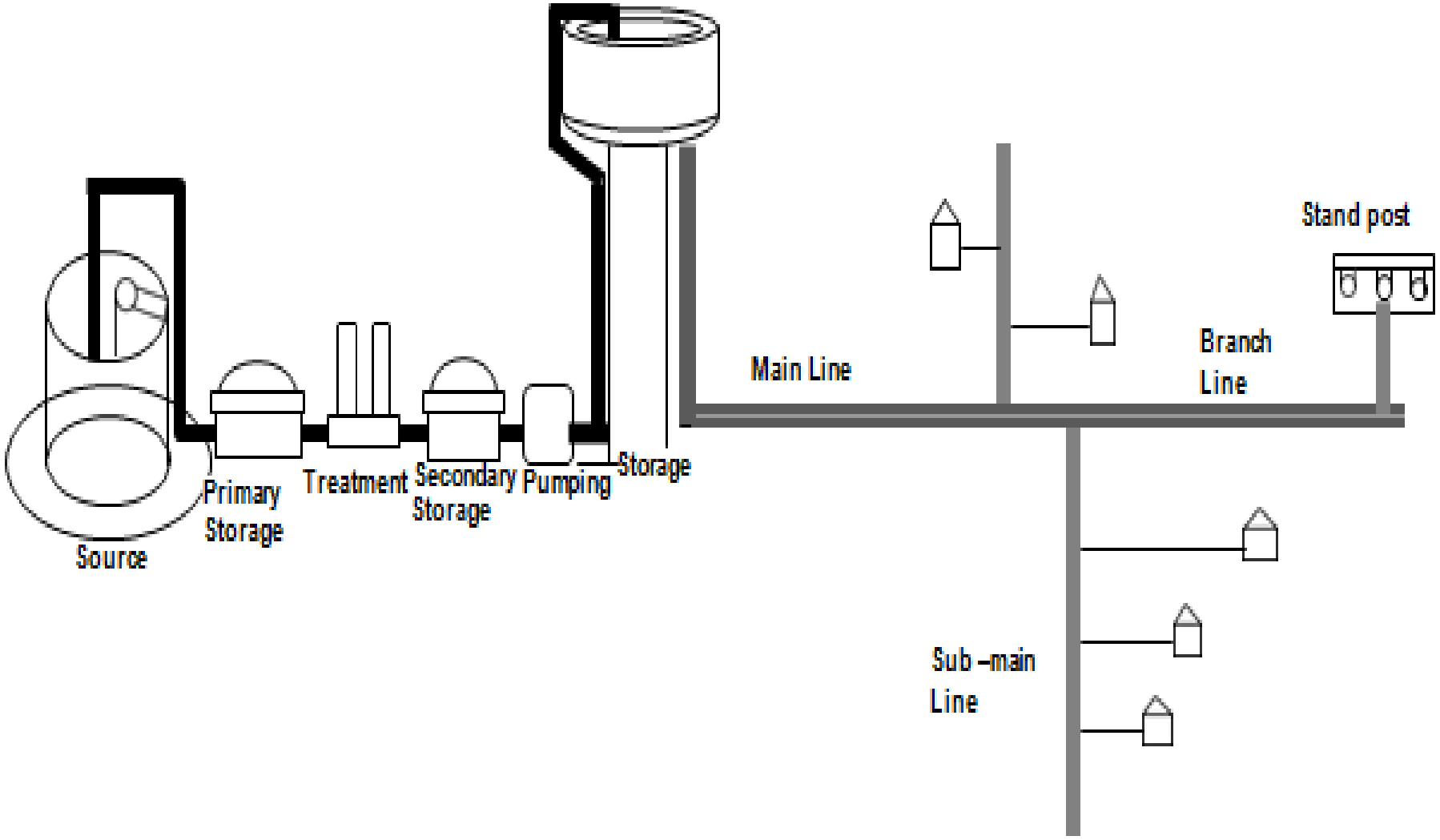
Typical Village/town water supply system constitutes of a gravity/pumping based transmission and distribution system from local/distant water source with needed water treatment system.

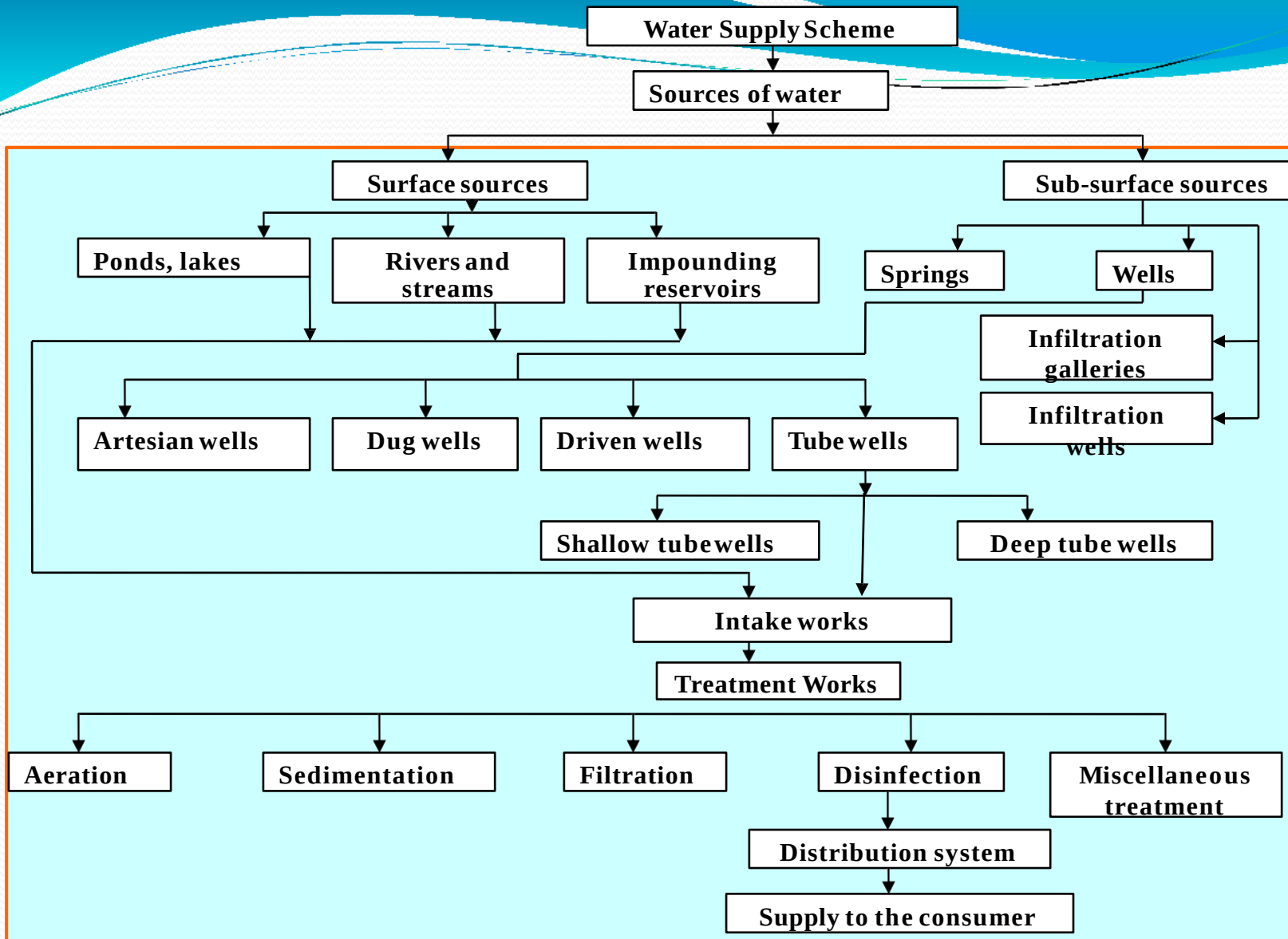




Standard Water supply System in village/town

Sources	Open Well, Tube Well, Hand pump, Pond, Dam Site, External Pipe Supply, Rain Water Harvesting System/Tank
Village/town level Treatment	Reverse Osmosis System (RO), Chlorination, Sedimentation, Sand Filter, etc.
Storage	Elevated Surface Reservoirs (ESR), Ground Service Reservoirs (GSR), Sump
Distribution	Main Line, Sub-Main Line, Branch Pipe Line, Household Level Tape, Stand Post, Washing Unit.

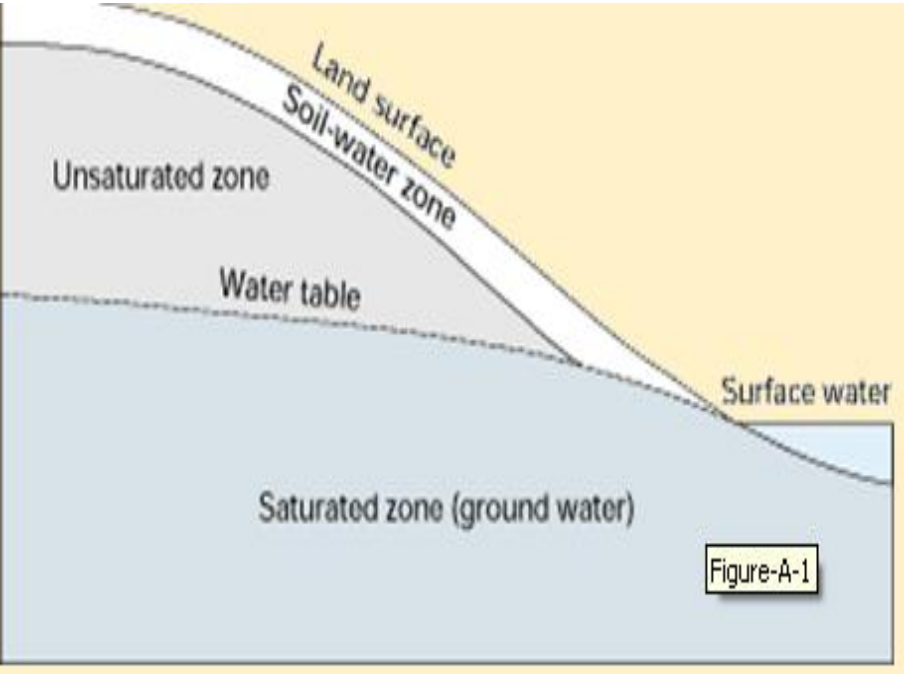

Source
Primary Storage
Treatment
Secondary Storage
Pumping
Storage
Main Line
Branch Line
Stand post
Sub –main Line

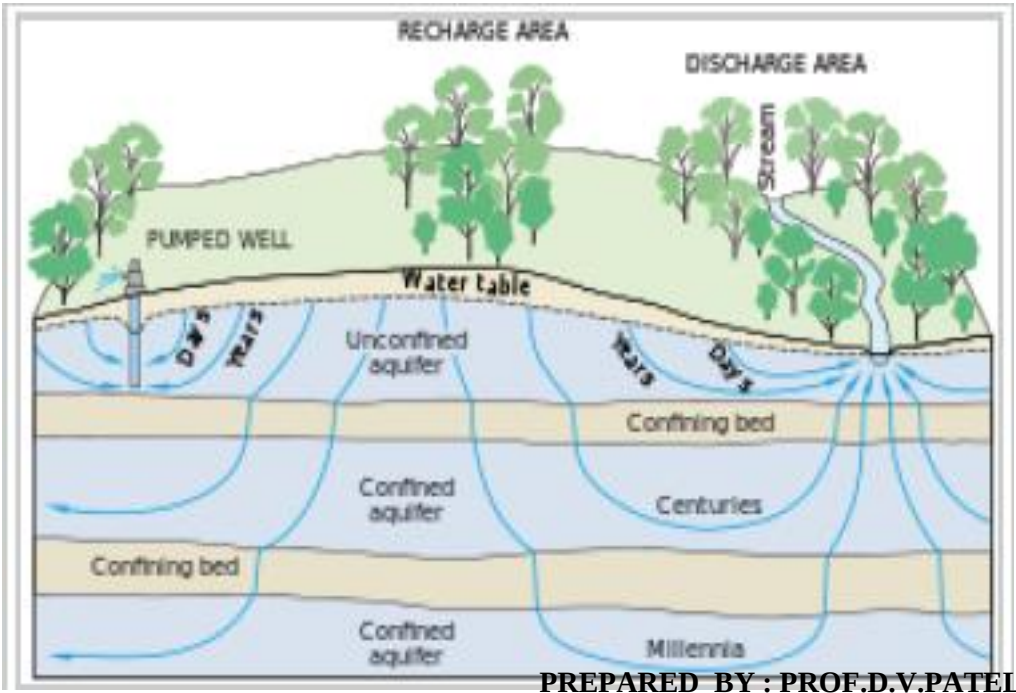


Schematic diagram of typical water supply scheme

Sources of Water Groundwater

- **Primary source of drinking water**
- Groundwater is the fresh water located in the pore space of soil and rocks.
- It is also water that is flowing within aquifers below the water table.
- The natural input to sub-surface water is seepage from surface water.
- The natural outputs from sub-surface water are springs and seepage to the oceans.











Lake Chungará and Parinacota volcano in northern Chile

- Surface water is water in a river, lake or fresh water wetland.
- Surface water is naturally replenished by precipitation and naturally lost through discharge to the oceans, evaporation, evapotranspiration and sub-surface seepage.

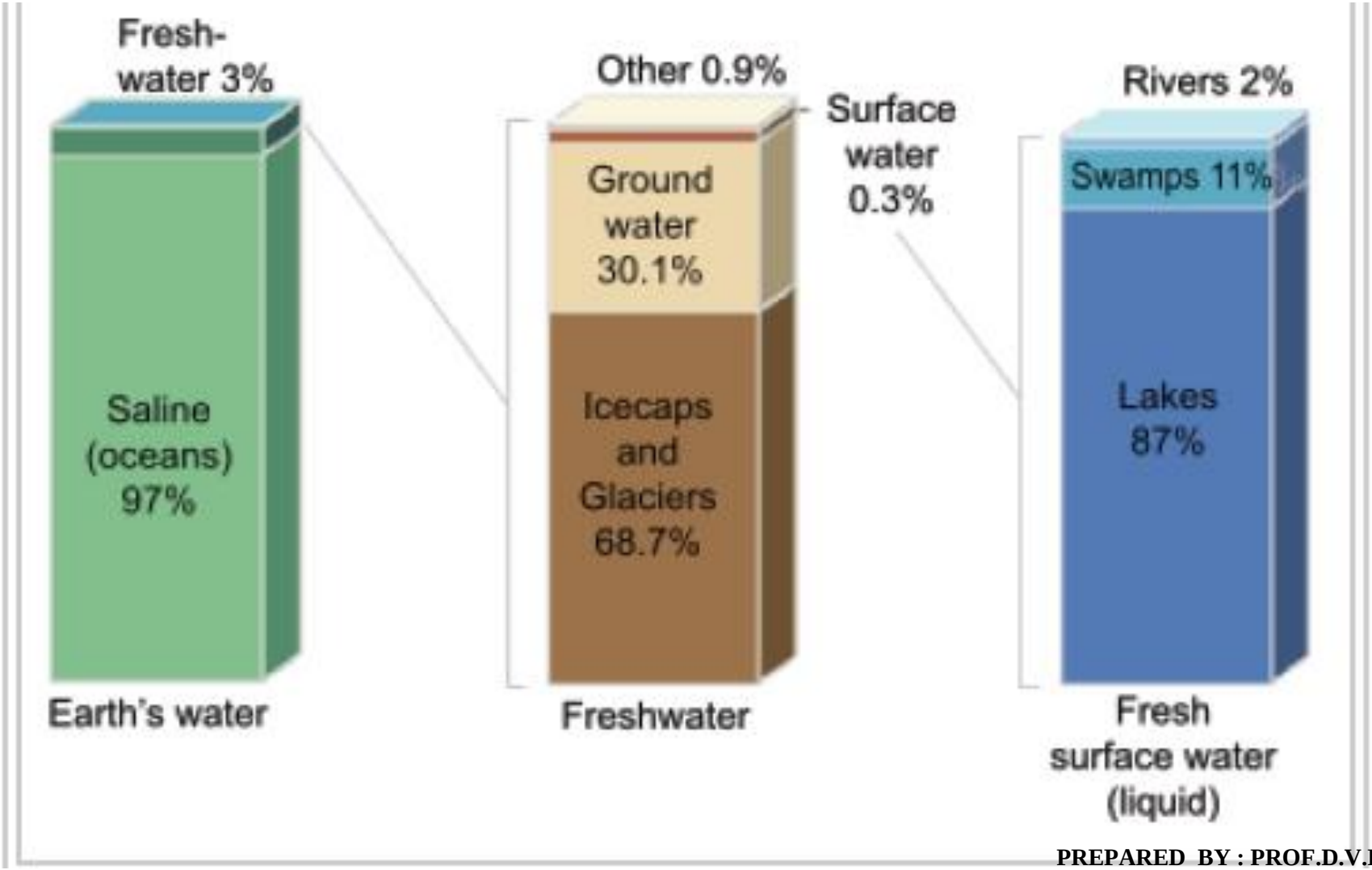


The total quantity of water in the system at any given time is dependent on many factors like:

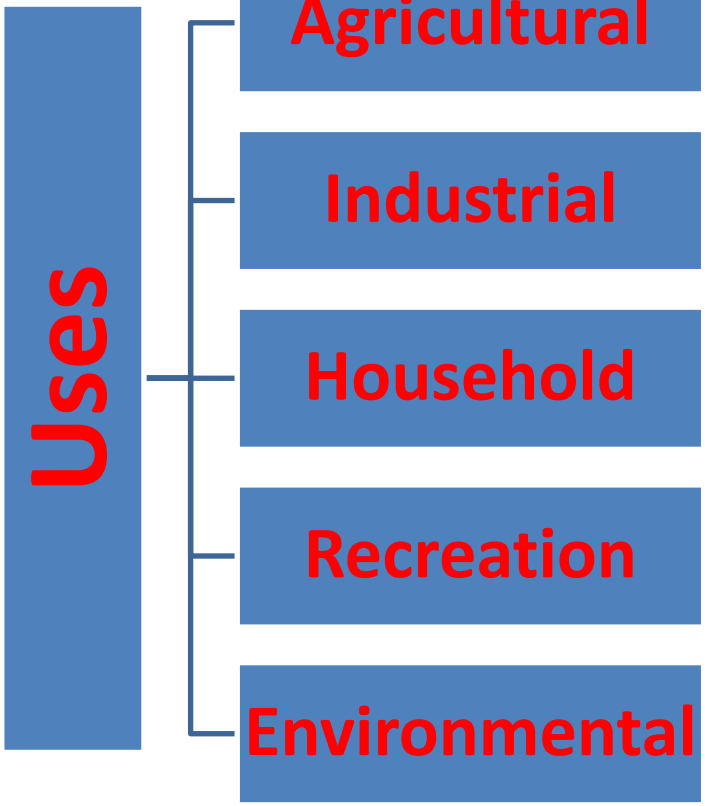
- Storage capacity in lakes, wetlands and artificial reservoirs,
- The permeability of the soil beneath these storage bodies,
- The runoff characteristics of the land in the watershed,
- The timing of the precipitation and local evaporation rates.



Distribution of Earth's Water



Uses of Water



70% of worldwide water used is for IRRIGATION, with 15-35% of irrigation withdrawals being unsustainable
22% of worldwide water is used in INDUSTRY. Major industrial users include Hydroelectric Dams, Thermoelectric Power Plants, which use water for cooling, ore and oil refineries, which use water in chemical processes, and manufacturing plants, which use water as a solvent
8% of worldwide water use is for HOUSEHOLD purposes. These include drinking water, bathing, cooking, sanitation, and gardening.
RECREATIONAL water use is usually a very small but growing percentage of total water use. Golf Courses, Hockey Grounds etc. are often targeted as using excessive amounts of water, especially in drier regions.
ENVIRONMENTAL water usage includes watering of natural or artificial wetlands, artificial lakes intended to create wildlife habitat, fish ladders, and water releases from reservoirs timed to help fish spawn, or to restore more natural flow regimes

Water Supply Schemes

- Before a project of water supply for a particular locality is taken in hand, a scheme is drawn out the different aspects of the whole scheme are carefully viewed from different angles.
- Water supply schemes are prepared by the combination of field observations and office work.
- Following are the points of importance in any water supply scheme:
  - Financial aspect
  - Population
  - Quality of water
  - Rate of consumption
  - Sanitary survey of the area
  - Sources of water supply
  - Topography of the area
  - Trends of town development

Water Supply Schemes

- **Financial aspect:** The data regarding the availability of funds for the fulfillment of the water supply scheme should be obtained in the initial stage of the scheme only. The scheme should then be adjusted in relation to the funds available. Every step should be taken to make the scheme as economical as possible.
- **Population:** From the available census of previous years, the present population should be determined and it is general practice to make the scheme to accommodate population after three or four decades. Failure to provide for future expansion results in great hardship in future.
- **Quality of Water:** The more pure water (Source of supply) is, the less is the cost of treatment. Hence, samples of available sources of water should be taken and properly analyzed and the results of various tests should be thoroughly studied to suggest an economical water supply scheme.
- **Rate of consumption:** The demand of water depends on various uses such as domestic, industrial, public etc. the rate of consumption per capita should be decided by carefully considering all the possible uses. This rate, when multiplied by the population, give the total quantity of water required for the water supply scheme.

Water Supply Schemes

- **Sanitary survey of the area:** The sanitary of area surrounding the available water sources should be carefully carried out. Such a survey helps in estimating the possible pollution or contamination of water from such sources.
- **Sources of water supply:** It is quite clear that the success of a water supply scheme entirely depends on a good source of supply of water. The source should be selected while keeping in view its adequacy, quality of water and cheapness.
- **Topography of the area:** The topography map of the area to be served by the scheme should be prepared and it should be suitable in relation to low lying area, ridges, density of population, etc. The study is essential to develop a simple & economical water supply scheme.
- **Trends of town development:** The trends of town development in future should be predicted & properly adjusted in the water supply scheme. Such trend may take various forms such as possibility of new industries, public recreation, public institutions, residential blocks, etc.

The period of design for important components of water supply Projects

S.No:	Component	Period of Design In years
01	Clear water service reservoirs	15
02	Conveying pipes for raw water and clear water	30
03	Distribution system	30
04	Electrical Motors and pumps	15
05	Infiltration works	30
06	Storage reservoirs	50
07	Water treatment units	15

WATER DEMAND

- While planning a water supply scheme, it is necessary to find out not only the total yearly water demand but also to assess the required average rates of flow and the variations in these rates. The following quantities are therefore generally assessed and recorded.
- Total annual volume in Litres or million litres. (1MLD = 10^6L/d)
- Annual Average rate of flow in litres per day , i.e. V/365
- Annual average rate of flow in litre per day per person (l/c/d), called **per capita demand**.

CLASSES OF WATER DEMAND

- **Domestic demand**

  This includes the water required in private building for drinking ,bathing, gardening, sanitary purpose, etc.

  As per IS : 200 l/c/d (with fully flushing system)

  135 l/c/d (for weaker sections and LIG)

- **Public demand**

  it represents the water demand for public utility purpose.

  like washing of public parks, gardening, washing on roads, public fountain. 20 l/c/d

- **Industrial demand**

  It represents the water demand of industries which are earlier existing or are likely to be started in future.

  As per IS : 50 l/c/d (for normal industries)

  450 l/c/d (industrial cities)

CLASSES OF WATER DEMAND

- **Commercial demand**

  Water requirement for institutions , hotels , schools, colleges, offices.

  As per IS : 20 l/c/d (for normal commercial area)

  50 l/c/d (highly commercial area)

- **Fire demand**

  in populated or industrial area fires generally breakout and may lead to serious problem. For control that situation require sufficient quantity water that is called fire demand.

- **Water demand require for thefts and wastes**

  This includes the water lost in leakage and stolen water due to unauthorised water connection.

Water Demand

1. ***Domestic Water Demand:***

   In this case water is needed for drinking, bathing, washing, utensils & cleaning of house & sewer flushing etc. domestic demand to be taken as 50% of Total water requirement.

2. ***Industrial or commercial Water Demand:***

   It includes water for factories, hotels, markets etc. the consumption depends upon the type of industries & the city where it is located. The percentage of water required varies from 20 to 25 % of Total water requirement.

3. ***Public Water Demand:***

   In this case water is needed for colleges, schools, Hostels, public gardens, picnic points, large hospitals, city community halls, jails etc. the percentage of water requirement for this purpose to be taken as 10% of Total water requirement.

4. ***Fire Water Demand:***

   It is very important for the municipal body to give adequate protection against fire, during fire the demand of water is very high because large quantity of water is required on urgent basis to control on the fire for this purpose fire hydrant are provided on all street at some extant from where water can be pumped at the rate of 950 to 1150 Liters / minute.

Following formulas are used to calculate fire demand.

Fire Water Demand:

I) Kuchling's formula:

$$Q = 3182 \sqrt{P}$$

Where as

Q = Total water quantity (Liters/minute)

P = Population of the town in thousands



II) Free man's Formula:

$$Q = 1136 [(P/5)+10]$$

II) National Board of Fire Formula:

$$Q = 4637 \sqrt{P}(1-0.01 \sqrt{P})$$

Note:

about 1 to 5 L/C/d may be taken as fire demand. It depends on factors like

possibility of fire, nature of town habits of peoples etc.

Losses & Wastages:

About 10 to 15% of Total water requirement goes for losses & wastages dueto unauthorized connection & leakages.

Variation in Water Demands

- The total per capita average demand in Pakistan about 200 to 300 L/C/d, where as in USA it is about 500 to 700 L/C/d.
- In any community water demands varies, depending upon factors like
  climate, duration of water supply, standard of living, drainage system.
- For example water consumption is more in morning time and evening time it varies hourly, daily.
- Water demand more on Sunday due to leave while planning any water supply scheme we have to calculate the average per capita demand & depending on that the Total quantity of water is determined for the Town, City, Village.
- It is important to note that mostly water supply schemes are designed
  for the period of 15 to 30 years.

PER CAPITA DEMAND

- It is the annual average amount of daily water required by one person and includes the domestic use, industrial and commercial uses, public use, wastes etc. It may therefore be expressed as:

- It includes the domestic use, industrial and commercial use, public use, wastes and thefts, fire demand etc.

Per capita demand(q) in l/d/h or l/c/d =

$$\frac{\text{Total yearly water requirement of the city in litres (V)}}{365 \times \text{Design population}}$$

Factors affecting ‘per capita demand’

1. Social and economical status of consumer
2. Cost of water
3. Climatic conditions
4. Industrial and commercial activities
5. Pressure in distribution system
6. Size of the city
7. Types of habitat of people
8. Quality of water supplies
9. Development of sewage facilities

* Types of Water Demand — Water demand (lpcd)

(IS 1172 : 1957)

1) Domestic Water Demand — 135 (135 to 200)
2) Industrial water Demand — 50
3) Institutional & Commercial — 20
4) Demand for public use — 10
5) Fire demand — (a) Freeman for. $Q = 1136\left[\frac{P}{5}+10\right]$
   (b) Kuichling for. $Q = 3182\sqrt{P}$
   (c) Buston for. $Q = 5663\sqrt{P}$
   ↳ $Q = 4637\sqrt{P}\left[1-0.01\sqrt{P}\right]$ — P — in thousand, Q — in lpm
   (d) National board of Underwriters formulae. (fire)
6) Demand to compensate Losses & theft — W.S. system 55 lpcd [for well maint. & full metered] extra provided = 20% total consumptn
7) Total water demand — 270 to 335

(For partly M & P. tab 50%.)

* Variation in Water Demand

Total Avg. daily water demand (Q) = population * per capita water demand

*max^m seasonal consumptn = 1.3 * Annual avg. daily rate of demand.

PREPARED BY : PROF.D.V.PATEL

① Maximum daily water demand = 1.8 * Avg. daily water demand.

$\therefore \; Q_{max} = 1.8Q$

② Maxm hourly water demand = 2.7 * Avg. daily water demand

= 1.5 * maxm daily

$(Q_{max})_{Hourly} = 2.7Q$

③ Coincident draft:

C.D. = Max daily + Fire demand

④ Total Demand = C.D (or) total Draft $(Q)_{max\ hrly}$ whichever is greater.

Design period

Design period may be defined as:

"It is the number of years in future for which the given facility is available to meet the demand."

Or

"The number of years in future for

•which supply will be more than demand."

- Water supply projects includes permanent constructions at Intakes, treatment plants and overhead tanks and laying of pipe lines
- They can not be replaced or resized or reconstructed easily in every year, as per the needs of the increased population, as it include huge amount of capital

Design period contd..

- The design periods for various project components may be taken as follows

Component	Design period
Storage reservoir	50 years
Infiltration works	30 years
Pipe connections	30 years
Distribution systems	30 years
Water treatment units	15 years
Electric motors and pumps	15 years

Factors affecting Design Period

Selection of an appropriate design period for a particular facility / component of water supply system is very important and dependent upon following factors:

1. Length or Life of Structures.
2. Ease of Extension.
3. First Cost
4. Economy of Scale
5. Lead Time

Factors affecting Design Period

1. Length or Life of Structure:

- Pumps
- Conduits (Type of Pipes)

Type of Pipe	Average Service Life
Steel Pipes	25 – 50 Years
Cast Iron Pipes	100 Years
Concrete Pipes	75 Years

Factors affecting Design Period

2. Ease of Extension:

This factor affects design period of whole Water Supply System depending upon the easiness with which its components can be enlarged i.e. effort with which Capacity of the components can be increased.

e.g.

- Tubewell : Easy to drill.
- Treatment Plant : Easy to enlarge.

Factors affecting Design Period

3. First Cost:

In case of High first cost and High Rate of interest, design period should be SHORT, so that the total cost of project should not be distributed to present population only.

e.g. Houses

Factors affecting Design Period

4. Economy of Scale:

Economy of Scale means that per unit production cost of a facility decreases with the increase in scale (size) of facility.

4. <u>Economy of Scale:</u> (Contd…)

Example:

- 200mm diameter W/S pipe, 1km Long serves 3000 Persons.

  Cost = Rs. 300,000 =>

  Cost/Head = Rs. 100

- 400mm diameter W/S pipe, 1km Long serves 12000 Persons.

  Cost = Rs. 480,000 => Cost/Head = Rs. 40

Factors affecting Design Period

5. Lead Time:

- It is the time from the perception of the project to its start of operation.

Lead Time = Time for Design of Project + Construction Time

- When lead time is MORE, MORE will be the design period.

Design Periods and Design Flows

for Various Water Supply System Components

1. <u>Large Dams, Transmission Main (Conduit)</u>

These are COSTLY and difficult to enlarge.

Design Period: 25 to 50 Years

Reservoir: Design Capacities of Reservoirs should be adequate to provide MAXIMUM DAILY DEMAND

Conduit: Generally designed for MAXIMUM DAILY DEMAND at the end of Design Period.

2. TUBE WELLS:

These are easy to drill.

- Design Period: 5 Years

➢ Designed at PEAK HOURLY DEMAND, if NO Storage is to be provided.

➢ Designed at MAX.DAILY DEMAND, if storage is to be provided.

3. **Water Treatment Plant:**

- Easy to enlarge.
- Design Period: 10 – 15 Years.
- Designed for MAXIMUM Daily Flow.

4. **Pumping Station:**

- Easy to enlarge.
- Design Period: 10 Years.

5. **Distribution System:**

- Difficult to REPLACE.
- Design Period: 25 Years.
- Designed for:

➢ MAX. DAY DEMAND + FIRE DEMAND.

➢ PEAK HOURLY DEMAND.

Whichever is greater.

Why Design period is provided ?

Design period is provided because

- It is very difficult or impossible to provide frequent extension.
- It is cheaper to provide a single large unit rather to construct a number of small units.
- Too Frequent extensions of the system are not possible.
- Reasonable economies of scale are achieved

Future Water Requirements

Depend upon:

1. Selection of PER CAPITA
2. Water Consumption
3. Future Population Forecast
4. Design Period

Requirements of Good Distribution System

- Sufficient pressure head at all points
- The minimum residual pressure for a single storey building is
- 0.07 N/mm^2 and for a double storey building is 0.12 N/mm^2
- Requisite amount of water for fire fighting
- Economical
- Simple and easy to operate
- Distribution pipes should not be laid below sewer lines, to avoid pollution
- Min. pipe sizes of 100 mm for towns having population up to 50,000 and 150 mm pipes for those above 50,000

Systems Of Distribution

- GRAVITY OR DIRECT SYSTEM
- PUMPING OR INDIRECT SYSTEM
- COMBINED OR DUAL SYSTEM

Systems of distribution contd..

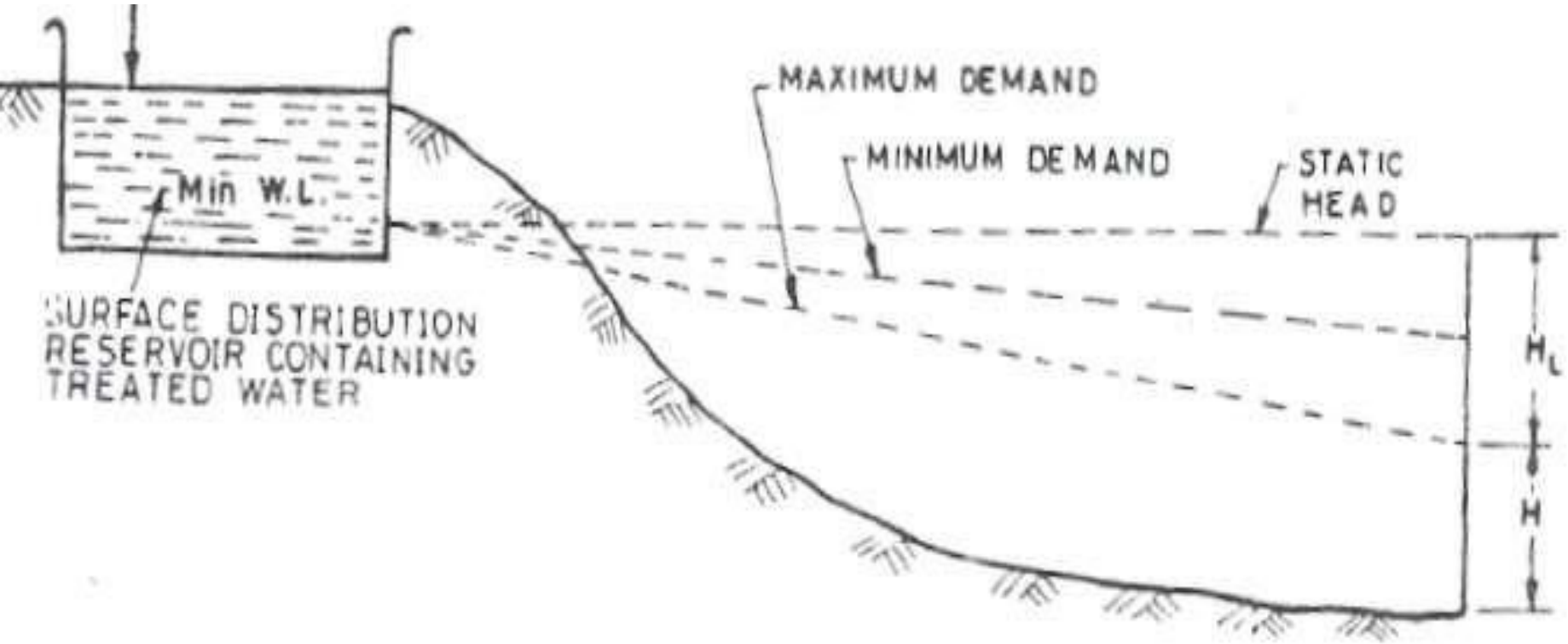


Fig. 1 GRAVITY SYSTEM

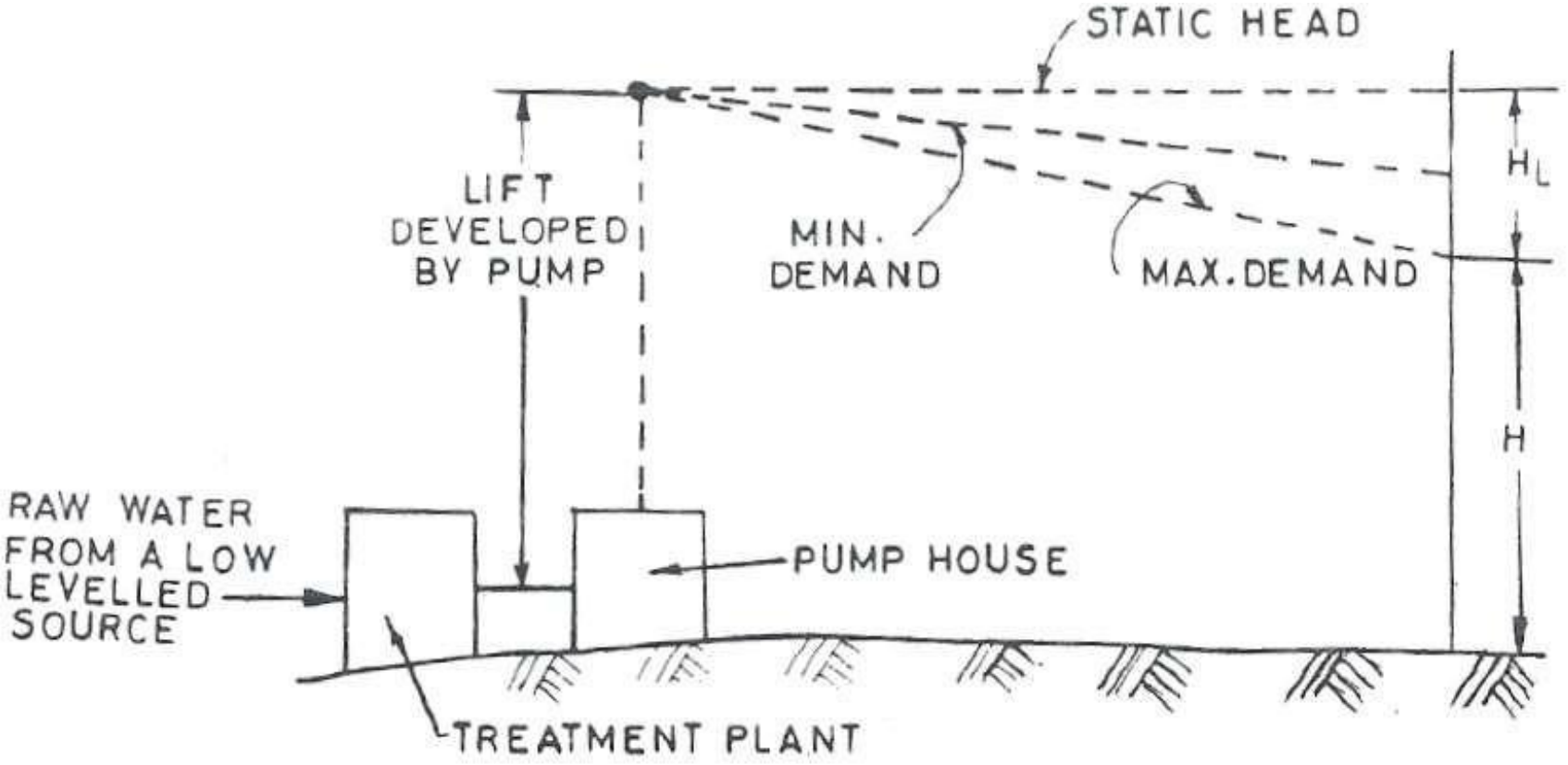


Fig.2 PUMPING SYSTEM

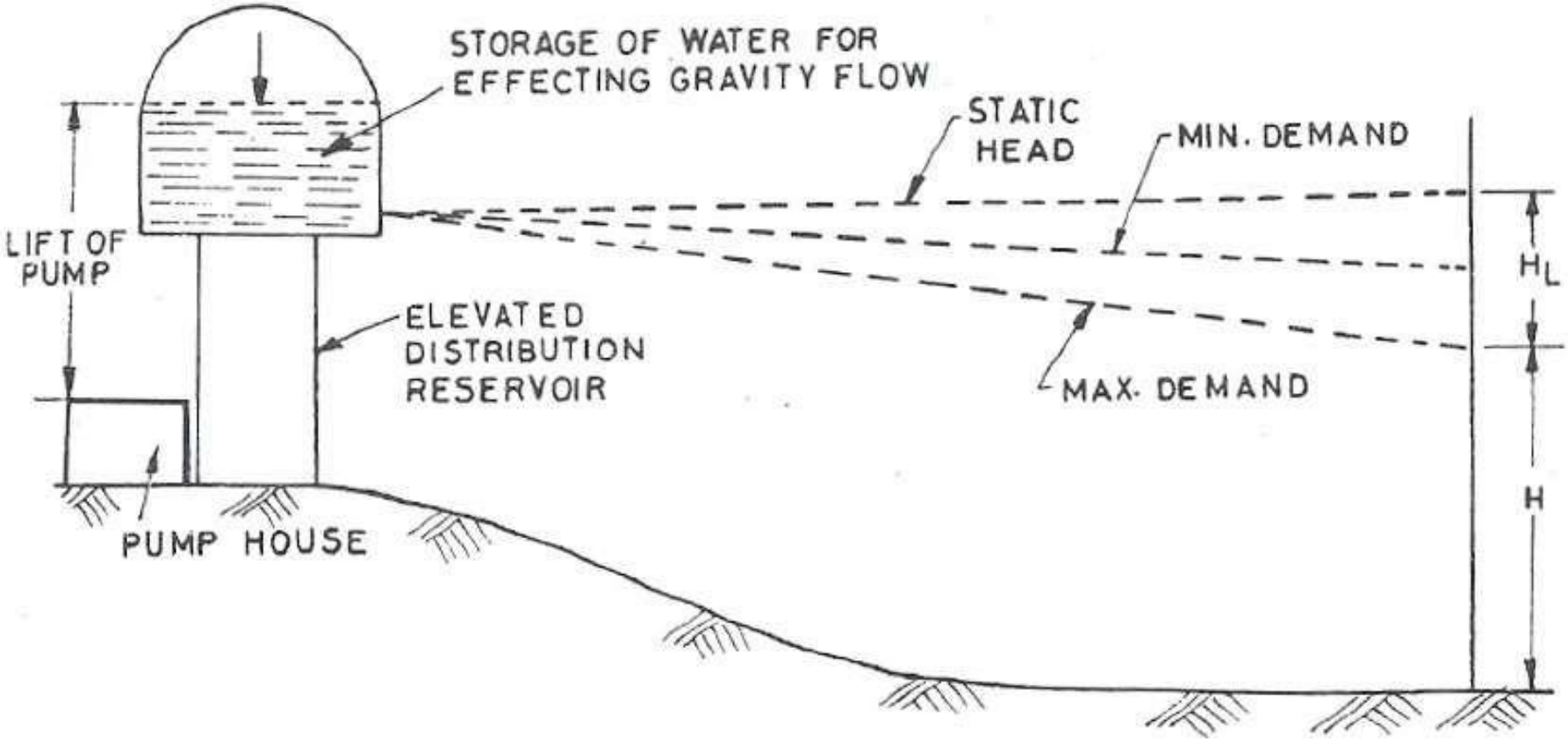


FIG.3 COMBINED GRAVITY & PUMPING SYSTEM

Estimating Population

- The term population is used to indicate the total number of human beings in a certain area at any particular time.
- The present population is obtained by referring to the statistics of census records prepared by the local body.
- The water supply projects is not designed only for present population. But it is made to accommodate the future population.
- The growth of population may be sharp, slow or even stationary depending upon the factors contributing to the future development of the locality such as coming up of new industries, trade expansion, etc.
- The future period for which various services units of water supply or sanitary engineering are designed is known as the *period of design.*
- The period of design may varies from 20 to 40 years or even upto 50 years.

POPULATION FORECASTING

Design of water supply and sanitation scheme is based on the projected population of a particular city, estimated for the design period.

Any underestimated value will make system inadequate for the purpose intended; similarly overestimated value will make it costly. Changes in the population of the city over the years occur, and the system should be designed taking into account of the population at the end of the design period.

Factors affecting changes in population are:

- increase due to births
- decrease due to deaths
- increase/ decrease due to migration
- increase due to annexation.

The present and past population record for the city can be obtained from the census population records. After collecting these population figures, the population at the end of design period is predicted using various methods as suitable for that city considering the growth pattern followed by the city.

POPULATION FORECASTING METHODS

- Arithmetic Increase method
- Geometric Increase Method
- Incremental Increase Method
- Decrease Rate of Increase Method
- Simple Graphical Method
- Comparitive Graphical Method